INCH-POUND
MIL-DTL-62749A
29 April 2010
SUPERSEDING
MIL-DTL-62749
28 August 1995

DETAIL SPECIFICATION

BRIDGE, FIXED: ALUMINUM, HIGHWAY, UNITED KINGDOM, MEDIUM GIRDER AND ERECTION, AND TRANSPORTATION EQUIPMENT, FULLY COMPETITIVE COMPONENTS FOR

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers all the fully competitive components for the bridge, fixed: aluminum, highway, United Kingdom, medium girder and erection, and transportation equipment which are listed in the Appendix to this specification.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in section 3, 4, and 5 of this specification. This section does not include documents in other sections of this specifications or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirement documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research Development and Engineering Center, ATTN: RDTA-EN/DM/STND/TRANS, MS# 268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mill/online/.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-53039 - Coating, Aliphatic Polyurethane, Single Component Chemical Agent Resistant.
MIL-DTL-64159 - Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-889 - Dissimilar Metals.

FEDERAL STANDARDS

FED-STD-595/34094 - Green 383

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or https://assist.daps.dla.mil/online/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract (see 6.2).

DRAWINGS

As listed in Appendix A.

(Copies of these documents are available from dami_standardization@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research Development, and Engineering Center, ATTN RDTA-EN/STND/TRANS, MS #268, 6501 E 11 Mile Road, Warren, MI 48397-5000)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017 or website www.asme.org.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedure and Tables for Inspection by Attributes

(Copies may be obtained from American National Standards Institute, 25 West 43rd Street, New York, NY 10036 (or website: http://www.ansi.org) or American Society for Quality, 600 North Plankton Avenue, Milwaukee, WI 53203 (or website: http://www.asq.org/))

AMERICAN WELDING SOCIETY (AWS)

AWS-A5.1	-	Covered Carbon Steel Arc Welding Electrodes.
AWS-A5.10	-	Bare Aluminum and Aluminum Alloy Welding Electrodes and Rods.
AWS-A5.18	-	Carbon Steel Filler Metals for Gas Shielded Arc Welding.
AWS-Dl.1	-	Structural Welding Code - Steel.
AWS-Dl.2	-	Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, (AWS), 550 N.W LeJeune Road, Miami, FL 33126 or website: http://www.aws.org)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC, AGENT

National Motor Freight Traffic Association Rules.

(Copies may be obtained from American Trucking Association Inc, ATTN: Traffic Order Section, 1001 North Fairfax Street, Suite 600, Alexandria, VA 22314 or website: http://nmfta.org.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules

(Copies may be obtained from Railinc, 7001 Weston Parkway, Suite 200, Cary, NC 27513 or website: https://www.railinc.com/)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Description. All the fully competitive components of the bridge erection and transportation equipment, hereinafter known as the fully competitive components, shall be as specified herein (see Appendix).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances are permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available on request, for inspection by the contracting officer or the designated representative. The dimensions shown on the drawings apply at temperatures from 68 to 70°F and dimensional limits and surface texture designations apply after plating and before painting.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor, and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The fully competitive components shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the components may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with fully competitive component(s) and shall make information available upon request to the contracting officer or his designated representative.

3.3.2 Recovered material. The components, pieces and parts incorporated in the fully competitive components shall be newly fabricated from recovered materials to the maximum extent practicable, provided the components produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the fully competitive components.

3.4 Welding electrodes.

3.4.1 For steel. Welding electrodes for steel shall conform to AWS A5.1 and A5.18 classification E60XX or E70XX series, size as required.

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3.4.2 For aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, classification ER 4043, size as required.

3.5 Performance. The components shall perform as a system without evidence of the following defects when tested in accordance with 4.5.2:

a. Malfunctioning of components.
b. Binding of parts during launching, erection and disassembly.
c. Permanent distortions.
d. Material defects, such as cracks, and open joints and weld deterioration.

3.6 Treatment and painting. The portions of the fully competitive components, normally painted, internal and external, shall be cleaned, treated, and painted. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383, #34094 of FED-STD-595 conforming to MIL-DTL-64159 or MIL-DTL-53039.

3.7 Identification markings. The components shall be identified in accordance with MIL-STD-130 and as specified in the applicable drawings.

3.8 Zinc coating. Zinc coating shall be as specified on the applicable drawings.

3.9 Inspection equipment (gauges). Inspection equipment, when required for verifying dimensions of fully competitive components of the erection and transportation equipment, shall be furnished by the contractor. Gauges shall be used to verify interchangeability by gauging interfacing dimensions. Gauges shall have been calibrated and certified by the contractor prior to production of any component and shall be calibrated and certified as required during the course of the contract.

3.10 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves and of adjoining parts, except for adjustment, and without selection for fit or performance.

3.11 Proof and preload. The contractor shall apply proof and preloads to the fully competitive components as specified on applicable drawings.

3.12 Repair of fully competitive components. Rejected fully competitive components may be repaired and submitted for inspection. Each repaired component shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection. Defective welds shall be removed by mechanical methods, and the repair welding may be accomplished with the material at any temper. Bending, forming, and straightening of repaired components shall be done as specified on applicable drawings.

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3.13 Workmanship.

3.13.1 Steel fabrication. Steel may be cut by shearing, sawing or flame cutting. Burned surfaces of flame cut material shall be ground or machined to remove cooling checks. All bends shall be made by use of metal dies or fixtures to insure uniformity in size and shape. All burrs shall be removed.

3.13.2 Aluminum fabrication. Aluminum used in the fabrication of the component shall be free from kinks and sharp bends. Straightening of material shall be done by methods that will not cause injury to the metal. Surfaces and edges of all finished members shall be smooth and free from kinks, bends and winds. All forms of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines. Aluminum shall not be cut by methods using heat. Aluminum less than 3/8 inch thick may be cut by shearing; aluminum 3/8 inch thick and over shall be cut by routing or by high speed saws. Shearing and sawing shall be done neatly and accurately. Material used must comply with the original material mechanical properties.

3.13.3 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. All fasteners shall be correctly torqued according to the specifying document and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.13.4 Welders and welding.

3.13.4.1 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural welding Code - Steel.
AWS D1.2 - Structural Welding Code - Aluminum.
ASME Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and welding and Brazing Operators.

3.13.4.2 Welding. The surfaces of parts to be welded shall be free from dirt, scale, paint, grease and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal arc (GMAW, which is also called metal insert gas or MIG). All welded parts shall be free from slag, flux, welded spatter, cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wire brushed clean. Tack welds shall be subject to the same quality requirements as final welds. Procedure and sequence shall be such that distortion and shrinking will be held to a minimum.

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3.13.4.2.1 Steel welding. Where appropriate, welding shall be carried out to welding procedures which conform to AWS D1.1.

3.13.4.2.2 Aluminum welding. When welding Aluminum Alloy, the metal temperature of aluminum members to be welded shall not be less than +5°C. Aluminum-magnesium alloys containing 4.0 to 5.5% magnesium, such as 5083, 5086, and 5456 should not be preheated to more than 93°C prior to welding. For other wrought aluminum alloys, preheating temperature should not exceed 149°C. Welds may be ground, filed, buffed or chipper and shall be free from unacceptable discontinuities, in accordance with the relevant specifications. Welding procedures shall be established on samples simulating actual material to be welded. The welding procedures shall include shielding minimum, manual per automatic travel speed, base material, preheat temperature, thickness, heat treated condition, joint preparation, position, and electrode. All welding of aluminum alloy components shall be in accordance with the required specification. Welds must be deposited by the metal inert gas (MIG) welding process, using shaved electrode wire conforming to AWS A5.10, classification ER 4043. In all other respects welding shall conform to AWS D1.2.

3.13.4.2.3 Welding fixtures. Shop fabricated components (except for minor parts) shall be assembled in fixtures or frames and welded while held securely in position. Minor parts may be held together by means that will insure secure and proper positioning. The fixtures or frames shall be designed to minimize distortion of the parts being welded.

3.13.5 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein, and on the applicable drawings as shown.

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article examination. If specified (see 6.2) one first article component, will be selected at random by the Government from the first batch of each type fully competitive component produced from production tooling and will be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.2.2 First article tests. If specified (see 6.2) each first article fully competitive component shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection. In addition to any test specified as part of the first article test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the first article tests and failure of such additional tests shall have the same effect as failure of those tests specified as first article tests.

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4.2.3 Inspection failure. Failure of any first article fully competitive component to meet any requirement specified herein during and as a result of the examination and tests specified in 4.5 shall be cause for rejection of that first article fully competitive component and shall be cause for refusal by the Government to continue acceptance of production of that type of fully competitive component until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on that type of fully competitive component(s) previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that, that type of fully competitive component(s) accepted prior to the completion of first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.3 Conformance inspection.

4.3.1 Sampling. Sampling shall be in accordance with ANSI/ASQC Z1.4.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in (applicable test method document or applicable paragraphs) the specification. The inspection and final acceptance shall be accomplished by a designated government Quality Assurance Representative (QAR) or Defense Contract Management Agency (DCMA) agent.

4.5 Inspection procedure.

4.5.1 Examination. Unless otherwise specified (see 6.2), contractor inspection equipment shall be used to inspect dimensions. The fully competitive components shall be examined as specified herein for the following defects.

4.5.2 Tests. If specified (see 6.2), each of the first article fully competitive components will be incorporated in Government furnished MGB sets and used in conjunction with the other MGB components, and providing they perform in accordance with the requirements of the operating manuals, the Government will authorize acceptance.

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TABLE I. Classification of defects.

Category	Defect	Method of examination
Critical:	None	
Major:		
101	Dimensions not as specified (see 3.1.1)	
102	Materials are not as specified (see 3.3)	
103	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).	
104	Dissimilar metals as defined in MIL_STD_889 are not effectively insulated from each other (see 3.3.1.1).	
105	Used, rebuilt or remanufactured components, pieces or parts incorporated in the fully competitive components (see 3.3.2).	
106	Proofloading not completed or not as specified (see 3.11).	
107	Cleaning, treatment, and painting not as specified (see 3.6).	
108	Zinc coating missing or not as specified (see 3.6).	
109	Welding and welds not as specified (see 3.13.4).	
110	Welding and welding operators not qualified as specified (see 3.3.13.1).	
111	Workmanship not as specified (see 3.13).	
Minor:		
201	Stenciling and identification marking not as specified (see 3.7).	

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The fully competitive items are intended to be used in conjunction with other components when transporting, erecting or retrieving the bridge, fixed, aluminum, highway, United Kingdom, medium girder. The erection equipment is capable of erecting and

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retrieving bridges spanning gaps of 30 to 162 feet. The transportation equipment is to be used when the bridge is in the transportation mode and provides an efficient means of moving the bridge and erection components from one bridge site to another.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this purchase description.
b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Which fully competitive items, identified (by drawing number) are required (see 1.1), and the quantity of each.
d. Time frame required for submission of the first article component(s) and number of each applicable type of component required (see 3.2).
e. If the Government requires any of the first article examination and tests (see 3.2).
f. If top coat color is other than specified (see 3.6).
g. Packaging requirements (see 5.1).
h. Any special marking required (see 5.2).

6.3 Subject term (key words) listing.

Assembly
Construction
Structure
Tool Kit

6.4 Definitions.

6.4.1 Recovered materials. "Recovered materials" means materials that have collected or recovered from solid waste (see 6.4.2).

6.4.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material resulting from industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations. section 23.402).

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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LIST OF COMPETITIVE ITEMS

	Description	NSN	Drawing No.
1.	Hammer, hand		13229E6798
2.	Wrench, ratchet, reversible		13229E6799
3.	Footwalk		13229E6801
4.	Bearer, footwalk		13229E6802
5.	Post, footwalk		13229E6803
6.	Rope, guard		13229E6804
7.	Marker, bridge, guide assembly		13229E6805
8.	Lug, tie-down		13229E6810
9.	Bracket, adapter, tie-down		13229E6811
10.	Strap assembly 5,000 lb		13229E6812
11.	Strap assembly 10,000 lb		13229E6813
12.	Sling		13229E6814
13.	Bag, equipment		13229E6815
14.	Tie-down, cargo		13229E6816
15.	Bar assembly, launching nose, push		13220E6822
16.	Push bar assembly, launching long		13229E6826
17.	Cable extractor		13229E6830
18.	Clip, retainer		13229E6831
19.	Rear bumper assembly		13229E6832
20.	Adapter, push bar		13229E6839
21.	Adapter, pallet to trailer		13229E6840
22.	Basket assembly, equipment		13229E6841
23.	Aid assembly, panel erection		13229E6847
24.	Pallet assembly		13229E6871

Custodians:
Army – AT
Navy - MC
Air Force – 99

Preparing Activity:
Army AT

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